

The Cathedral of Cause and Experience

An essay on the construction of Heaven based on a reliable rule

The Productive Refusal

There is a peculiar strength in saying "no". This is not the reactive "no" of fear or defensiveness. It is the principled refusal to accept what cannot be grounded. In mathematics, this refusal has taken the form of a small but persistent skepticism toward completed infinities: the idea that an actually finished infinite totality can exist as a whole. Mainstream mathematics, as taught in virtually every university, freely accepts such totalities as everyday tools; the skeptic is the exception, not the rule. In physics and philosophy, similar suspicions arise whenever theories appeal to structures so vast, so unconstrained, that they cease to explain anything at all. This essay explores a bold and, at first glance, unlikely thesis: that by rejecting completed infinities, we can construct a more reliable foundation not only for physics and consciousness, but for ethics—and ultimately, for something we may dare to call Heaven.

Heaven here is not a supernatural realm handed down from above. It is something built—carefully, rigorously, and joyfully—from the most de-

pendable ingredients we can find. It is a structure of experience, sustained across time, in which happiness is not fragile, not fleeting, but robust. To build such a structure, we need three pillars: a theory of reality, a theory of experience, and a principle of value. The proposal is that the Wolfram Physics Project can serve as the first pillar; Tononi's Integrated Information Theory can serve as the second; and from these, the third can be inferred: an ethics grounded in happiness and suffering, constrained by the preservation of an open future.

But everything begins with a refusal: the rejection of completed infinities. Without that, the rest dissolves into vagueness. With it, something surprisingly solid emerges.

The Illusion of Totality

Completed infinities are seductive. They promise completeness, closure, and a kind of omniscient perspective. In set theory, they appear as fully formed infinite sets. In cosmology, as entire multiverses. In metaphysics, they surface as the totality of Platonic forms: a timeless realm in which all possible structures exist in perfect abstraction. These ideas aim to capture everything—but in doing so, they risk explaining nothing.

The problem is not that infinity is large, but that a completed infinity is static: given in full, without construction, without the step-by-step unfolding that makes something intelligible. When everything exists simultaneously, nothing leads meaningfully to anything else. Causality blurs, expla-

nation turns circular, reliability vanishes.

If we are to build anything—especially something as delicate as a theory of Heaven—we need a foundation that is not merely vast, but dependable. Dependability requires that each part arises from something prior in a way that can be followed, reproduced, and trusted. A completed infinity offers none of this. It is an answer without a question, a totality without a process.

Rejecting completed infinities leaves open whether they exist in reality, because the alternative (creation from nothing) is equally strange. Either way, we can build without a completed totality. We embrace infinity in its incomplete forms: the future, the potential complexity, the ongoing size, and other unbounded aspects of reality.

Simple Rules, Complex Worlds

Stephen Wolfram's magnum opus, *A New Kind of Science* (NKS), rests on a deceptively simple discovery: extremely simple computational rules can generate astonishing complexity. From an initial condition, a single rule applies again and again, taking its own output as new input in an endless cascade of discrete steps. What emerges is not just chaos but intricate, structured behavior. These are patterns that resemble the fabric of physical reality itself.

Wolfram's proposal is radical yet elegant: the universe might be exactly

such a system. A hypergraph (a discrete network of relations) evolves step by step under a fixed rule. From this humble beginning, large-scale structures arise that behave like space, time, and causality. The arrow of time, the three-dimensionality of space, the unification of general relativity and quantum mechanics: all may be consequences of nothing more than a single rule running. The rule does not contain physics. Physics emerges from its repeated application.

This shifts our understanding of space and time fundamentally. They are no longer a static container or a pre-existing stage. Instead, space is constructed relationally by the hypergraph; time is the process of the rule being applied. Each moment follows from the previous one. The future is not already laid out, nor is it yet computed. Space itself grows and changes with every step. The universe is an ongoing computation, always in motion, always becoming.

Why One Rule Beats All

The Ruliad is a brilliant idea: all possible rules, all possible consequences, wrapped into one completed infinite totality. But if everything exists, nothing is uniquely explained. Causality requires exclusion: this happened because that happened, and not something else. A structure in which every alternative has already been realized loses the very essence of cause and effect.

The usual response is observer selection: we see this universe because ob-

servers like us have specific properties. But this becomes circular. The observer doing the selecting is itself a product of the very physics that needs explaining. The Ruliad merely hides the need for a starting point inside the observer. A single rule escapes this problem entirely. There is no pool of alternatives, no selection, no circularity. The rule simply runs, and we are what it produces.

This minimalism is its strength. There is nothing beneath the rule, nothing beyond it, nothing between its steps. Each state follows from the previous one by the same principle. Reliability without exception. And we have reason to believe we are not living in some exotic corner of an ultra-advanced universe. The pyramids were built by human cooperation, not by lost civilizations. Life evolved from a single ancestor through natural selection, not through intelligent design. The Big Bang was the first move, not a branch on an infinite tree. Around us are automatic processes, not hidden magic. A single unwavering rule excludes the supernatural—and in that exclusion, it gives us everything we need to understand.

Where the Rule Becomes Feeling

A universe generated by a simple rule gives us a foundation for physics. But Heaven is not made of physics alone. It is made of experience. To understand how experience arises within a causal structure, we turn to Integrated Information Theory (IIT), developed by Giulio Tononi.

IIT begins with a simple but powerful observation: every experience is a

unitary whole. You cannot split the experience of a rose into an "experience of red" and an "experience of shape" that exist independently. The experience is integrated. At the same time, every experience is specific: it is this rose, in this light, in this moment. A physical system that generates such an experience must therefore consist of parts that are both highly differentiated and deeply interdependent. IIT quantifies this with a measure called Φ (Phi): the degree to which a system's causal structure constitutes an irreducible whole, greater than the sum of its parts. The higher the Φ , the richer and more unified the experience.

This raises an immediate question: where does experience begin? If consciousness cannot magically appear from non-conscious matter, and cannot merely imitate experience without truly having it, then it must be a fundamental feature of reality. IIT implies that any system with nonzero Φ has some form of experience, a conclusion that aligns with panpsychism. At the micro level, the distinction between physical process and experience has not yet arisen. This state lies outside any recognizable experience—it is the irreducible precondition of subjectivity. IIT then tells us which causal structures stack these elementary transitions into something we recognize as experience: intrinsic, composed, informative, integrated, exclusive. This gives us a bridge between physics and consciousness.

Wolfram's computational hypergraph is not merely a mathematical abstraction; the transformations performed by the rule directly create a structure of real causal relations. IIT then provides the key to understanding what happens when this structure forms certain patterns: consciousness is identical to a local, maximally integrated causal structure. The

consciousness of a simple, self-contained system is a faint spark, barely integrated. The consciousness of a human brain is a galaxy of interdependence, achieving a momentary Φ through the incredible complexity of its internal connections.

Yet this does not yet explain why our consciousness is sometimes painful. The answer lies in evolution. Biological life, starting from a single common ancestor, was initially a method for maintaining and replicating complex patterns. Over billions of years, the networks became larger, more complex, and more conscious. But this awakening was not designed for harmony. The first minds were crude instruments of survival, driven by feedback loops of hunger, fear, and aggression. Suffering was the inescapable shadow of the first steps toward integration.

Two Conditions of One System

From this origin follows a crucial distinction that frees us from the myth that we need to suffer in order to be happy.

Suffering is the experience of negative valence with sufficient intensity—a structure where learned responses signal threat, fragmentation, or being overwhelmed. This suffering requires that Φ is high enough to feel that signal clearly. Happiness is the mirror: positive valence at sufficient intensity, with learned responses signaling harmony, safety, and the opportunity for ever-deepening complexity. Higher Φ deepens the experience and makes the valence feel more vivid and intense, whether pleasant or

painful. What determines the sign of that valence is the direction of Pavlovian conditioning, which unfolds through historical time. Φ sets the depth; conditioning sets the sign. Step by step we can habituate to anything, but habituation to patterns without negative side effects opens up the infinite future. Experiencing a symphony, feeling a deep connection, grasping an elegant truth—all of this is happiness, not merely because it activates a reward signal, but because that signal is conditioned toward things that do not exhaust themselves.

Evolution optimized us for survival, never for happiness. The human psyche is full of tragic shortcuts inherited from ages of scarcity. These are substances or behaviors that hijack the conditioning signal (valence) without increasing complexity. They mimic rich happiness but eventually destroy it. On the other hand, power structures—corporations, governments, ideologies—do not exist to make human lives better. They survive or perish by their own internal dynamics: growth, competition, replication. A political system that improves well-being may flourish, but only if that well-being also serves the system's self-preservation. When the two conflict, power preserves itself first. Most human suffering in modern society is not an accident of evolution. It is the waste product of systems that were never designed for us at all.

In the end, happiness requires no suffering as its contrast agent. This means it is possible to get used to forms of happiness without negative side effects, which is what we would need if the infinite future is to become completely positive.

Getting Used to What Never Gets Boring

At the heart of this vision lies a subtle but essential idea: genuine happiness can be based entirely on becoming attuned to things that never bore. This may sound paradoxical. How can habituation lead anywhere other than dullness?

The answer lies in the distinction between repetition and familiarity. Repetition leads to boredom because it offers nothing new. Familiarity, on the other hand, can coexist with novelty. One can become attuned to a pattern, a style, a way of engaging—while still encountering new variations within it. Depth and newness are not opposites.

Consider music. A great composition can be listened to many times without losing its power. Each listening reveals new details, new connections, new emotional resonances. The listener becomes more familiar with the work, but the experience does not become empty—it deepens. The piece has not changed, but the listener's capacity for integration has grown.

In a universe generated by a complexity-producing rule, this kind of depth is structurally guaranteed. There are always new patterns, new configurations, new forms of integration to be discovered. Because the rule ensures that novelty never runs out, the skill of attuned engagement can sustain happiness indefinitely. We can always become more sensitive, more creative, more deeply integrated with what the unfolding universe presents. There is no final plateau, no ultimate saturation—only an ever-receding horizon of richer harmony and freedom.

Happiness, in this framework, is not a destination. It is a skill: the ability to resonate with the unfolding complexity of the world, to create it, and to find in that creation an experience that is both harmonious and inexhaustible.

The Scientific Path to the Spiritual Goal

If happiness is conditioned to be the experience of harmonious complexity, and suffering that of disintegration, then the mission of a civilization becomes clear. The goal is to construct open-ended safety—a state that tolerates maximum diversity without significant, negative disruption to Φ systems like humans and animals.

This sounds utopian, but within this model it is a logical possibility and an endless process. It is possible because the underlying rule is trustworthy. Everything that threatens us (an asteroid, a virus, a psychosis) is a pattern within the causal graph, never an intrinsic, malevolent force of reality. And every pattern, with sufficient knowledge and collective effort, can be understood, predicted, and eventually averted.

Technology, in the broadest sense, is the totality of our tools for understanding and purposefully reshaping the environment. Cooperation is the mechanism by which individuals combine their understanding and effort toward goals none could reach alone. The development of vaccines, the construction of flood defenses, the alignment of artificial intelligence—all are examples of this collective power. Safety is merely a technological and

moral project: the culmination of science (understanding the rule and its possibilities) and politics (harmonizing individual will with collective insight). This project already has a track record. Smallpox no longer claims lives—not managed, not treated but effectively removed from the world we inhabit. The causal pathway that once produced millions of deaths has been closed. Polio is following the same trajectory. These are not examples of suffering being reduced; they are examples of entire categories of suffering receding from the horizon of ordinary human experience. The scientific path runs through laboratories, through cooperation, through the patient accumulation of understanding. The spiritual goal is infinite happiness.

Once safety is achieved, a new domain of existence opens. Happiness begins to stack upon itself. Every new understanding, every work of art, every deepened relationship, every wondrous discovery adds a new layer of harmonious integration. There is no ceiling, because true complexity is an infinite domain of harmonies yet to be found. The same rule that generates physical complexity without limit is the foundation on which happiness can stack indefinitely. The scientific path to the spiritual goal does not end at the elimination of suffering—that is only the threshold. Beyond it lies the real construction: an open-ended accumulation of experience, each layer made possible by the inexhaustible novelty that a complexity-producing rule guarantees. What every religious tradition has called Heaven—and has placed beyond reach—turns out to be constructible by beings like us in a universe like ours.

The Paradox of Freedom and Responsibility

One specter remains: determinism. If the rule determines everything—from the Big Bang to this moment—what remains of freedom? Is the experience of choice an illusion? And if everything is determined, does that counsel passivity?

The answer is a firm no, and the reason illuminates something deeper about the structure of agency. It is a category error to confuse determinism with unfreedom at the level of the experiencing system. A stone has no freedom; its path is entirely determined by external forces. A conscious being is also determined, but the chain of causality now runs, to a very large extent, *through their own internal structure*. The brain models the world, weighs possibilities, and the outcome is the result of its own intrinsic dynamics. This internal self-causation is precisely what freedom means. A free being is one whose actions are primarily caused by their own integrated knowledge, values, and commitments.

Crucially, this freedom grows. In a primitive, suffering-afflicted state, the dynamics of a brain are dominated by a few crude external inputs: hunger, fear, cold. The space of genuine choice is minimal. Through technology, cooperation, and the construction of safety, we absorb these deterministic blows. Freedom grows, because the options we can genuinely consider, and the values we can integrate, continue to expand.

With this growing freedom comes proportionally growing responsibility. As the actions of humanity—driven by science and technology—achieve planetary and eventually cosmic reach, the importance of causal under-

standing increases proportionally. It is deeply irresponsible to build particle accelerators, reshape climates, or program artificial intelligence without a correspondingly deep understanding of their consequences for the complex, integrated systems that house consciousness. The spiritual goal is as important as the scientific path.

Constructing the Neverending Paradise

By rejecting completed infinities, we free ourselves from the illusion of totality and embrace the reality of construction. A simple rule, iterated over time, provides a sound foundation for physics. Integrated Information Theory connects that foundation to experience. From there an ethics emerges, grounded in the concrete reality of happiness and suffering, constrained by the preservation of an open future, and animated by the growing freedom of beings who understand their own causal nature.

Heaven, in this framework, is a direction rather than a destination. It is the ongoing cultivation of systems that can experience ever-deepening forms of joy, without exhausting the possibilities. If the universe really works as presented, it never stops generating novelty. It is sustained by the endless capacity for growth that a complexity-producing rule guarantees.

The image that remains is that of a cathedral of cause and experience. Reality is not an infinite museum of all possible worlds, nor a product of blind chance without meaning. It is a dynamic process, driven by a simple

rule that is reliable to its deepest core. From that rule springs a causal graph, and in the folds of that graph, local maxima of integrated information arise. That is us.

The first steps were painful. The initial state of the first complex mind contained as much suffering as awe. But the path is visible. We can reshape the crude, accidental patterns that survival generated into harmonious patterns we choose ourselves. We can outgrow the errors of destructive desires and political power games. We can construct ample safety—not by magic, but by science, compassion, and cooperation. And in that safety, happiness stacks upon itself, layer after layer, without a ceiling.

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